

The Expectation-Maximization Algorithm

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Introduction

Let's first recall what Maximum Likelihood means.

The goal of MLE is to find the parameter θ that makes the observed data most likely.

Suppose we have observed data $X = \{x_1, x_2, \dots, x_n\}$, and we *model* each x_i as an independent realization from a distribution family $p(x_i | \theta)$ parameterized by θ .

The likelihood function:

$$L(\theta | X) = \prod_{i=1}^n p(x_i | \theta)$$

The Maximum Likelihood Estimate (MLE):

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta | X)$$

We choose θ that makes the observed data most probable under our assumed model.

Why MLE Breaks Down with Latent Variables

In many real-world problems, we only observe the outcomes but not the hidden states that generate them.

For example, in financial markets we observe the daily VIX, but we do not directly observe which market regime (calm or turbulent) produced each value.

If the regime labels were known, computing the likelihood and applying MLE would be straightforward.

However, **when these latent states are unobserved:**

- the likelihood involves a sum over all possible hidden-state assignments,
- making direct maximization analytically intractable.

This is precisely the challenge in mixture models, regime-switching models, and many incomplete-data problems.

To address this challenge, we use the **The Expectation–Maximization (EM) algorithm**.

The EM Algorithm is a way to find a Maximum Likelihood Estimate for problems where we have missing data. Missing data can be understood as unobserved or latent variables which would allow us to solve for the MLE if we had them. If we think of data in a spreadsheet, we are usually missing a column of the dataset, not a row.

An Intuitive Example: The Two-Coins Problem

Suppose we want to learn the probability of getting heads for two coins, Coin A and Coin B. Let θ_A denote the probability of heads when flipping Coin A, and let θ_B denote the probability of heads when flipping Coin B.

We observe 4 sequences of coin flips.

HHTHH

HTTHT

HTHTH

TTTTT

It is very easy to estimate the probability of heads for each coin, since we know which observations are from which coin.

7 out of 10 flips from Coin A are heads.

$$\hat{\theta}_A = \frac{7}{10} = 0.7$$

2 out of 10 flips from Coin B are heads.

$$\hat{\theta}_B = \frac{2}{10} = 0.2$$

HHTHH

HTTHT

HTHTH

TTTTT

But what if we didn't know which filp came from which coin? It would be a lot harder to estimate θ_A and θ_B .

One solution might be to just guess which of the sequences belongs to each coin. (The first and the third sequence have more heads than tails, so we might guess they came from the same coin; the left two have more tails than heads, so we might guess they came from the same coin.)

This could work but what if we are wrong? Our estimates are going to be incorrect...

This would be especially difficult if there were many coins and many sequences.

Let's allow for the possibility for us to improve our estimates **if our initial guesses are incorrect**.

1. Guessing which sequence is from which coin, accounting for uncertainty.

This will be the **E-Step**. We will take an expectation, which accounts for the uncertainty.

2. Estimating θ_A and θ_B .

This is the **M-Step**. We will choose the $\hat{\theta}_A$ and $\hat{\theta}_B$ that maximize the likelihood of observing our data.

And we will alternate between these two steps until our estimates stop changing.

Heads	Tails	Likelihood with coin A	Likelihood with coin B
H	T	$(\hat{\theta}_A)^H(1 - \hat{\theta}_A)^T$	$(\hat{\theta}_B)^H(1 - \hat{\theta}_B)^T$
3	7	0.000075	0.001678
4	6	0.000175	0.000419
7	3	0.002224	0.0000066
9	1	0.012106	0.0000004

The initial guess is:

$$\hat{\theta}_A^{(1)} = 0.7 \qquad \hat{\theta}_B^{(1)} = 0.2$$

We compute the likelihood of each sequence under each of the two coins (the current guess of the coin's bias).

$$0.7^3 \times 0.3^7 = 0.000075$$

$$0.2^3 \times 0.8^7 = 0.001678$$

Coin B is about 19 times more likely. In a way, the probability of being from coin A is about 0.05, and from coin B is about 0.95.

We can convert the likelihoods into probabilities with Bayes' rule.

$$\begin{aligned} P(A | X_i, \hat{\theta}^{(t)}) &= \frac{P(X_i | \hat{\theta}_A^{(t)}) \times P(A)}{P(X_i | \hat{\theta}_A^{(t)}) \times P(A) + P(X_i | \hat{\theta}_B^{(t)}) \times P(B)} \\ &= \frac{P(X_i | \hat{\theta}_A^{(t)}) \times \frac{1}{2}}{P(X_i | \hat{\theta}_A^{(t)}) \times \frac{1}{2} + P(X_i | \hat{\theta}_B^{(t)}) \times \frac{1}{2}} \\ &= \frac{L(A)}{L(A) + L(B)} \end{aligned}$$

Heads	Tails	Likelihood with coin A	Likelihood with coin B	Prob(Coin A)	Prob(Coin B)
H	T	$(\hat{\theta}_A)^H(1 - \hat{\theta}_A)^T$	$(\hat{\theta}_B)^H(1 - \hat{\theta}_B)^T$	$\frac{L(A)}{L(A) + L(B)}$	$\frac{L(B)}{L(A) + L(B)}$
3	7	0.000075	0.001678	0.0428	0.9572
4	6	0.000175	0.000419	0.2944	0.7056
7	3	0.002224	0.0000066	0.997	0.003
9	1	0.012106	0.0000004	0.99997	0.00003

We compute the relative likelihoods for all the sequences, given the current set of parameters.

Attribute Heads and Tails to Each Coin

Heads	Tails	Likelihood with coin A	Likelihood with coin B	Prob (Coin A)	Prob (Coin B)	Heads/Tails Attributed to Coin A	Heads/Tails Attributed to Coin B
H	T	$(\hat{\theta}_A)^H(1 - \hat{\theta}_A)^T$	$(\hat{\theta}_B)^H(1 - \hat{\theta}_B)^T$	$\frac{L(A)}{L(A) + L(B)}$	$\frac{L(B)}{L(A) + L(B)}$	$\frac{H \times P(A)}{T \times P(A)}$	$\frac{H \times P(B)}{T \times P(B)}$
3	7	0.000075	0.001678	0.0428	0.9572	0.128H 0.3T	2.87H 6.7T
4	6	0.000175	0.000419	0.2944	0.7056	1.178H 1.767T	2.822H 4.233T
7	3	0.002224	0.0000066	0.997	0.003	6.979H 2.99T	0.02H 0.0088T
9	1	0.012106	0.0000004	0.99997	0.00003	8.9997H 0.99996T	0.0003H 0.0003T

Since there is only a 0.0428 probability that this sequence was generated by Coin A, we attribute 4.28% of both the 3 heads and the 7 tails to Coin A.

$$3H \times 0.0428 = 0.128H$$

$$7T \times 0.0428 = 0.300T$$

Next, we figure out how many heads and tails in all the sequences are attributed to Coin A and Coin B by summing the corresponding columns in the table.

Heads	Tails	Likelihood with coin A	Likelihood with coin B	Prob (Coin A)	Prob (Coin B)	Heads/Tails Attributed to Coin A	Heads/Tails Attributed to Coin B
H	T	$(\hat{\theta}_A)^H(1 - \hat{\theta}_A)^T$	$(\hat{\theta}_B)^H(1 - \hat{\theta}_B)^T$	$\frac{L(A)}{L(A) + L(B)}$	$\frac{L(B)}{L(A) + L(B)}$	$\frac{H \times P(A)}{T \times P(A)}$	$\frac{H \times P(B)}{T \times P(B)}$
3	7	0.000075	0.001678	0.0428	0.9572	0.128H 0.30T	2.87H 6.70T
4	6	0.000175	0.000419	0.2944	0.7056	1.178H 1.767T	2.822H 4.233T
7	3	0.002224	0.0000066	0.997	0.003	6.979H 2.997T	0.021H 0.008T
9	1	0.012106	0.0000004	0.99997	0.00003	8.999H 0.999T	0.0003H 0.00003T
Sum						17.285H 6.057T	5.715H 10.943T

With these totals, we can update our estimates of the parameters:

$$\hat{\theta}_A^{(2)} = \frac{17.285}{17.285 + 6.057} = 0.74$$

$$\hat{\theta}_B^{(2)} = \frac{5.715}{5.715 + 10.943} = 0.343$$

These values become our new estimates of θ_A and θ_B !

Next, we: (1) compute the likelihoods using the updated estimates, (2) convert them to relative probabilities, and (3) apportion heads and tails between the coins. This gives new totals for each coin under the current θ_A and θ_B .

Heads	Tails	Likelihood with coin A	Likelihood with coin B	Prob (Coin A)	Prob (Coin B)	Heads/Tails Attributed to Coin A	Heads/Tails Attributed to Coin B
H	T	$(\hat{\theta}_A)^H(1 - \hat{\theta}_A)^T$	$(\hat{\theta}_B)^H(1 - \hat{\theta}_B)^T$	$\frac{L(A)}{L(A) + L(B)}$	$\frac{L(B)}{L(A) + L(B)}$	$\frac{H \times P(A)}{T \times P(A)}$	$\frac{H \times P(B)}{T \times P(B)}$
3	7	0.000032	0.00213	0.015	0.985	0.044H 0.1T	2.96H 6.9T
4	6	0.00009	0.0011	0.076	0.924	0.30H 0.46T	3.70H 5.54T
7	3	0.002	0.000159	0.931	0.069	6.52H 2.79T	0.48H 0.21T
9	1	0.0174	0.00004	0.9975	0.0025	8.978H 0.998T	0.022H 0.024T
Sum						15.84H 4.35T	7.157H 12.649T

We then re-estimate the parameters and take these as the updated estimates.

$$\hat{\theta}_A^{(3)} = \frac{15.84}{15.84 + 4.35} = 0.785$$

$$\hat{\theta}_B^{(3)} = \frac{7.157}{7.157 + 12.649} = 0.361$$

We repeat this procedure over and over again. Eventually, the answers will stop changing very much. We reach a local optimum.

$$\hat{\theta}_A^{(10)} = 0.801$$

$$\hat{\theta}_B^{(10)} = 0.3715$$

So the question is...

What did we just do? How does it work in general? And why does it work?

Mathematical Foundations of the EM Algorithm

- The EM algorithm is a method for estimating parameters, similar to maximum likelihood.
- It works when there is some concept of **missing data**.
- In the coin flip example, we observed sequences of heads and tails, but we did not observe the labels of which sequences came from which coin.
- By introducing the missing variable (which coin the sequences came from), we actually make the problem easier.

Maximum Likelihood vs. EM Algorithm

Maximum Likelihood

(M algorithm)

- Choose the θ that maximizes the likelihood.
- We can do this because the likelihood is relatively simple (or at least we can write it down, since we have all the information).

EM Algorithm

- We want to choose the θ that maximizes the likelihood.
- This is hard because we have missing data.
- To find the θ that maximizes the likelihood:
 - We will try to fill in the missing data.
 - Then find the θ that maximizes the likelihood.
 - And repeat until we are happy with our answer (it stops changing).

EM Algorithm

- 1 Find the **Q function**, which is related to the log-likelihood function but is based on attempting to complete the missing data and accounting for uncertainty.
- 2 Maximize this Q function (which improves the likelihood too!).
- 3 Repeat over and over until our answer stops improving.

$$\begin{aligned}
 L(\theta) &= \log p(X \mid \theta) = \log \int_C p(X, C \mid \theta) dC \\
 &= \log \int_C p(C \mid \theta) dC
 \end{aligned}$$

We want to maximize the log-likelihood. However, it can be difficult to do directly.

We can make the problem easier by introducing two key ideas. One is the concept of missing or completed data.

X = the observed data

C = the complete data
(Contains both the original data X and the missing data.)

θ = the parameter we want to estimate

$$\begin{aligned}
L(\theta) &= \log p(x \mid \theta) = \log \int_C p(x, c \mid \theta) dc \\
&= \log \int_C p(c \mid \theta) dc \\
&= \log \int_C \frac{p(c \mid \theta)}{p(c \mid x, \hat{\theta}^{(t)})} p(c \mid x, \hat{\theta}^{(t)}) dc \\
&= \log \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\frac{p(c \mid \theta)}{p(c \mid x, \hat{\theta}^{(t)})} \right]
\end{aligned}$$

The other is the use of current estimates of the parameters.

X = the observed data

C = the complete data
(Contains both the original data X and the missing data.)

θ = the parameter we want to estimate

$\hat{\theta}^{(t)}$ = the estimate of the parameter θ at time t

$$\begin{aligned}
L(\theta) &= \log p(x \mid \theta) = \log \int_C p(x, c \mid \theta) dc \\
&= \log \int_C p(c \mid \theta) dx \\
&= \log \int_C \frac{p(c \mid \theta)}{p(c \mid x, \hat{\theta}^{(t)})} p(c \mid x, \hat{\theta}^{(t)}) dx \\
&= \log \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\frac{p(c \mid \theta)}{p(c \mid x, \hat{\theta}^{(t)})} \right] \\
&\geq \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\log \frac{p(c \mid \theta)}{p(c \mid x, \hat{\theta}^{(t)})} \right]
\end{aligned}$$

We can move the log inside the expectation using Jensen's inequality.

$$L(\theta) = \log p(x | \theta) = \log \int_C p(x, c | \theta) dc$$

$$= \log \int_C p(c | \theta) dc = \log \int_C \frac{p(c | \theta)}{p(c | x, \hat{\theta}^{(t)})} p(c | x, \hat{\theta}^{(t)}) dc$$

$$= \log \mathbb{E}_{c|x, \hat{\theta}^{(t)}} \left[\frac{p(c | \theta)}{p(c | x, \hat{\theta}^{(t)})} \right]$$

$$\geq \mathbb{E}_{c|x, \hat{\theta}^{(t)}} \left[\log \frac{p(c | \theta)}{p(c | x, \hat{\theta}^{(t)})} \right]$$

By properties of logarithms and expected values, we can separate this into two expected values.

$$= \mathbb{E}_{c|x, \hat{\theta}^{(t)}} [\log p(c | \theta)] + \mathbb{E}_{c|x, \hat{\theta}^{(t)}} [-\log p(c | x, \hat{\theta}^{(t)})]$$

$$= Q(\theta | \hat{\theta}^{(t)}) + h(c, x, \hat{\theta}^{(t)})$$

We call the first part the **Q function**, which is a function of θ and $\hat{\theta}^{(t)}$.

The h function is called the differential entropy. We don't care about it much, because it's not a function of θ .

The EM algorithm works by improving the Q function (because it's too hard to improve the likelihood), which we obtain by attempting to complete the missing data.

$$L(\theta) \geq Q(\theta | \hat{\theta}^{(t)}) + h(c, x, \hat{\theta}^{(t)}) \quad (1)$$

Why does improving Q improve L ?

At the value $\theta = \hat{\theta}^{(t)}$, Equation (1) is an equality:

$$L(\hat{\theta}^{(t)}) = Q(\hat{\theta}^{(t)} | \hat{\theta}^{(t)}) + h(c, x, \hat{\theta}^{(t)}) \quad (2)$$

By subtracting Equation (1) and (2), we have

$$L(\theta) - L(\hat{\theta}^{(t)}) \geq Q(\theta | \hat{\theta}^{(t)}) - Q(\hat{\theta}^{(t)} | \hat{\theta}^{(t)}).$$

Thus, when we improve the Q function we will improve the likelihood function by at least as much.

We do improve Q at each step. In the **M step**, we maximize the Q function:

$$\hat{\theta}^{(t+1)} = \arg \max_{\theta \in \Theta} Q(\theta \mid \hat{\theta}^{(t)})$$

Thus by definition, we have

$$Q(\hat{\theta}^{(t+1)} \mid \hat{\theta}^{(t)}) \geq Q(\hat{\theta}^{(t)} \mid \hat{\theta}^{(t)}).$$

And thus

$$L(\hat{\theta}^{(t+1)}) \geq L(\hat{\theta}^{(t)}).$$

ok, but where was Q in the example we just worked on,
and where did we maximize it?

$$Q(\theta \mid \hat{\theta}^{(t)})$$

$$= \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} [\log p(c \mid \theta)]$$

$$= \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\sum_{i=1}^4 \log (\theta^{x_i} (1 - \theta)^{10 - x_i}) \right]$$

$$= \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\sum_{i=1}^4 (x_i \log(\theta) + (10 - x_i) \log(1 - \theta)) \right]$$

This is the pmf of a binomial distribution where x_i is the number of heads in the i^{th} sequence. These θ s are either θ_A or θ_B , but we don't know yet. Keep an open mind.

Use properties of logs.

$$\begin{aligned}
& Q(\theta \mid \hat{\theta}^{(t)}) \\
&= \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\sum_{i=1}^4 (x_i \log(\theta) + (10 - x_i) \log(1 - \theta)) \right] \\
&= \sum_{\text{coin}=A}^B \left[\sum_{i=1}^4 (x_i \log(\theta) + (10 - x_i) \log(1 - \theta)) \right] \times p(\text{coin} \mid x, \hat{\theta}^{(t)})
\end{aligned}$$

An expected value is a sum of a quantity times a probability.

We want to sum over all completions of the data.

(Don't worry about these steps too much if you are having trouble following. It gets a little confusing with all the different types of θ s.)

$$\begin{aligned}
& Q(\theta \mid \hat{\theta}^{(t)}) \\
&= \mathbb{E}_{c \mid x, \hat{\theta}^{(t)}} \left[\sum_{i=1}^4 (x_i \log(\theta) + (10 - x_i) \log(1 - \theta)) \right] \\
&= \sum_{\text{coin}=A}^B \left[\sum_{i=1}^4 (x_i \log(\theta) + (10 - x_i) \log(1 - \theta)) \right] \times p(\text{coin} \mid x, \hat{\theta}^{(t)}) \\
&= \sum_{\text{coin}=A}^B \sum_{i=1}^4 (\underbrace{x_i p(\text{coin} \mid x, \hat{\theta}^{(t)})}_{\text{Attributed Heads sequence } i} \log(\theta) + \underbrace{(10 - x_i) p(\text{coin} \mid x, \hat{\theta}^{(t)})}_{\text{Attributed Tails sequence } i} \log(1 - \theta)) \\
&= \sum_{\text{coin}=A}^B \sum_{i=1}^4 [\text{Attributed Heads sequence } i \times \log(\theta) + \text{Attributed Tails sequence } i \times \log(1 - \theta)]
\end{aligned}$$

These quantities are simply the expected (attributed) numbers of heads and tails.

$$\begin{aligned}
 & Q(\theta \mid \hat{\theta}^{(t)}) \\
 &= \sum_{\text{coin}=A}^B \left[\text{Attributed Heads} \times \log(\theta_{\text{coin}}) + \text{Attributed Tails} \times \log(1 - \theta_{\text{coin}}) \right] \\
 &= H_A \times \log(\theta_A) + H_B \times \log(\theta_B) + T_A \times \log(1 - \theta_A) + T_B \times \log(1 - \theta_B)
 \end{aligned}$$

H_A and T_A are the total attributed heads and tails to coin A, and H_B and T_B are the total attributed heads and tails to coin B.

We completed the E step!

Then, we maximize the Q function by taking partial derivatives of $Q(\theta \mid \hat{\theta}^{(t)})$ with respect to θ_A and θ_B .

$$Q(\theta \mid \hat{\theta}^{(t)}) = H_A \log(\theta_A) + H_B \log(\theta_B) + T_A \log(1 - \theta_A) + T_B \log(1 - \theta_B)$$

$$\frac{\partial Q}{\partial \theta_A} = \frac{H_A}{\theta_A} - \frac{T_A}{1 - \theta_A} = 0$$

$$H_A(1 - \theta_A) - T_A \theta_A = 0$$

$$H_A - H_A \theta_A - T_A \theta_A = 0$$

$$H_A - \theta_A(H_A + T_A) = 0$$

$$\hat{\theta}_A = \frac{H_A}{H_A + T_A}$$

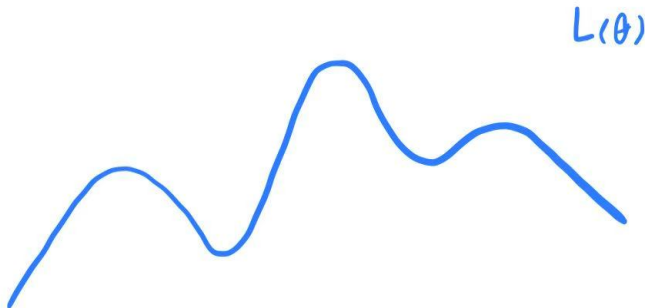
Similarly,

$$\hat{\theta}_B = \frac{H_B}{H_B + T_B}$$

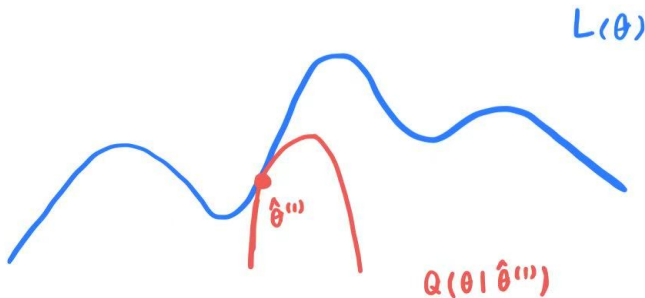
We completed the M step!

Visualizing the EM Algorithm

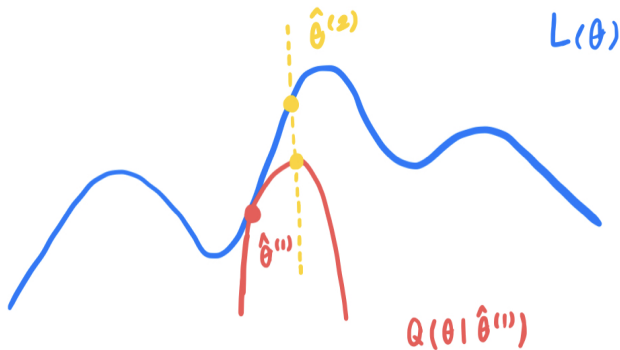
We want to maximize the likelihood function.



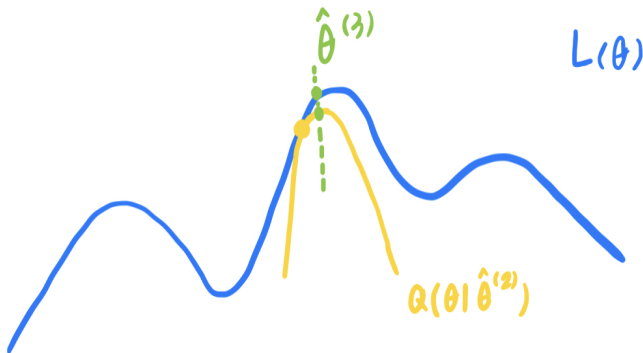
Here is the Q function, which is equal to the likelihood at one point, but below it everywhere else. Then we find the θ that maximizes Q .



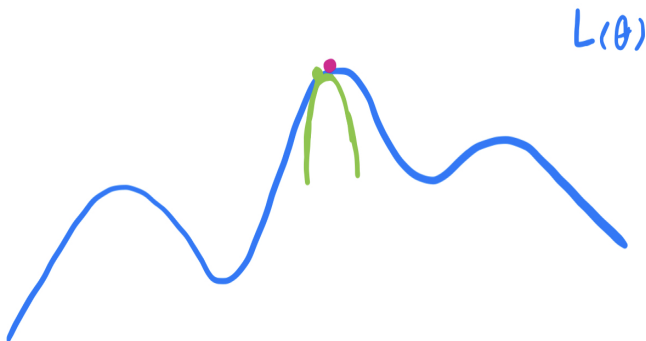
Here is our new estimate $\hat{\theta}^{(2)}$.



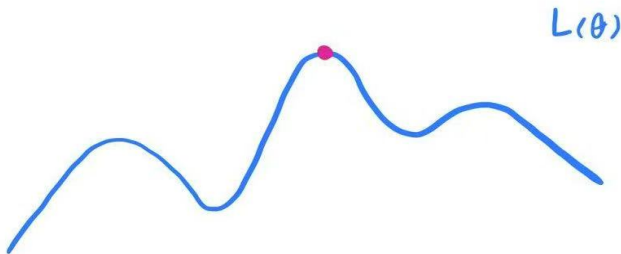
Let's draw a new Q function here.
Again $Q = L$ at the value $\theta = \hat{\theta}$.
Then, maximize Q again.



And repeat by creating another Q function.
Now, we maximize Q again.



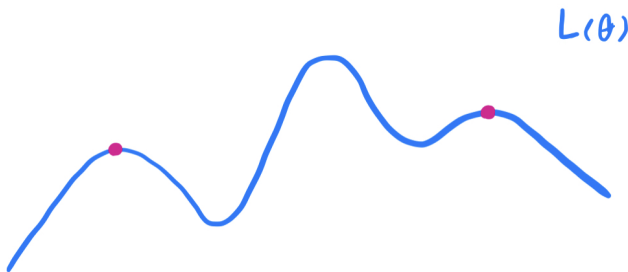
Eventually... we will reach the local maximum,
which is the global maximum here.



If we had started in a bad part of the likelihood function with our initial guess, we may have gotten trapped in a local maximum.

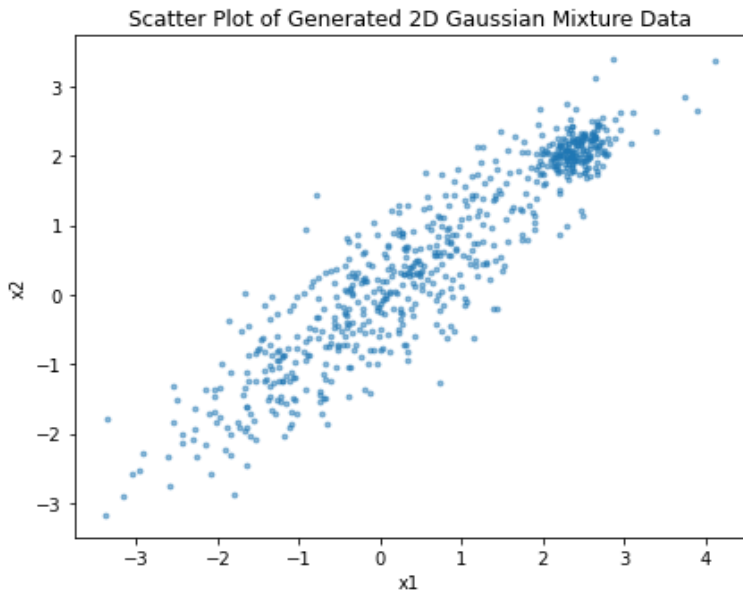
We can avoid this by choosing many different starting points.

If we run the EM algorithm from all these starting points, at least some will end up in the global maximum.

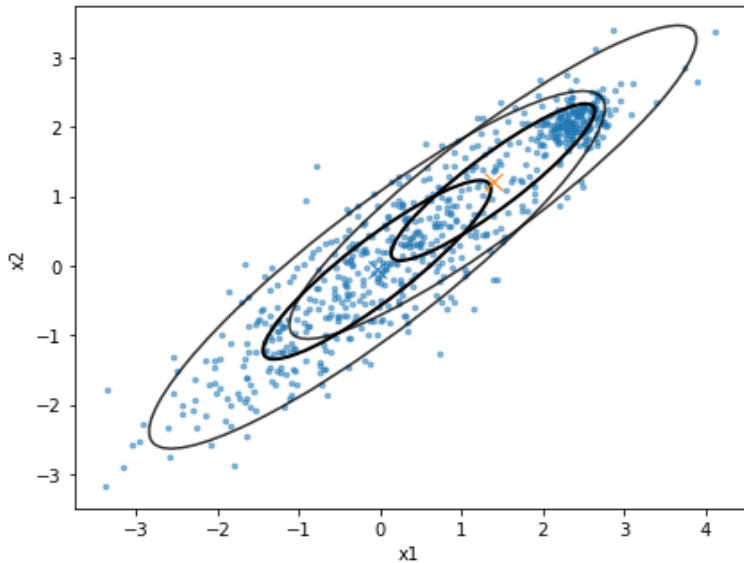


Fitting Gaussian Mixture Models with EM: A Financial Application

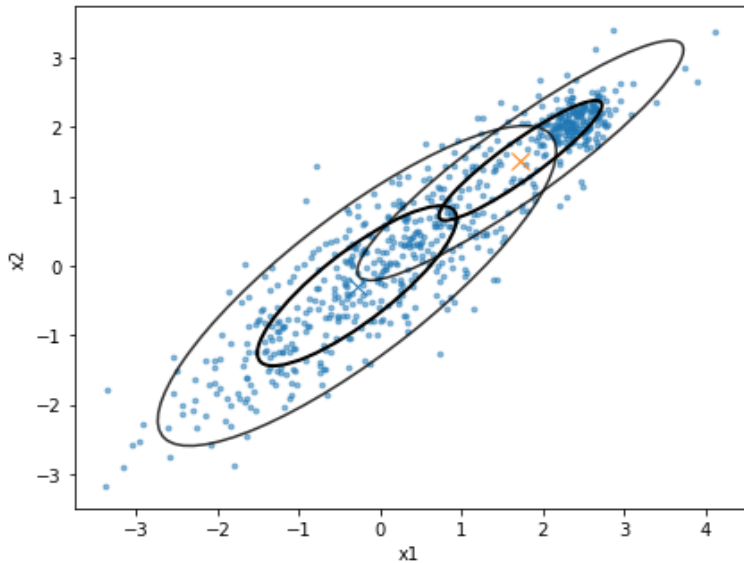
One common use of the EM algorithm is to fit the GMM.



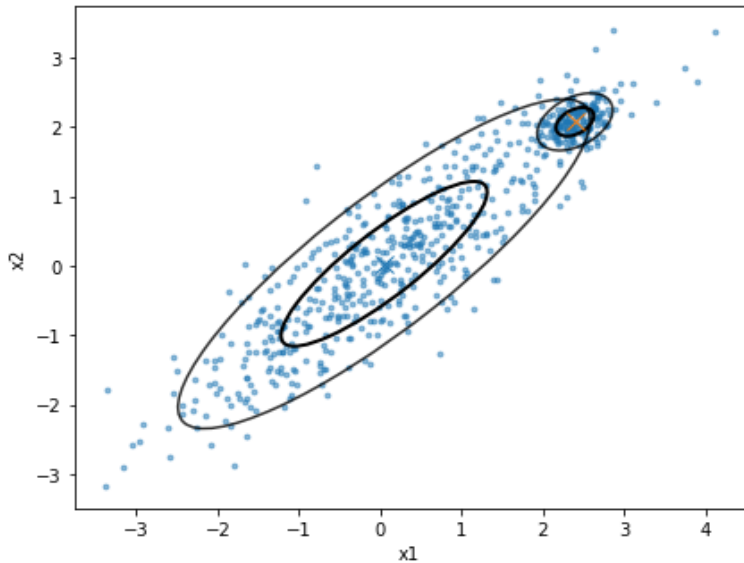
GMM via EM — After 1 Iteration(s)



GMM via EM — After 5 Iteration(s)



GMM via EM — After 20 Iteration(s)



The **CBOE Volatility Index (VIX)**—commonly referred to as the “Fear Index”—captures market expectations of near-term volatility and is widely interpreted as a measure of investor sentiment and risk aversion. Historically, the VIX exhibits clear regime-like behavior: during calm and optimistic market periods, it remains at low levels, while in times of financial stress or crises, it spikes sharply. This leads to a **multi-modal and fat-tailed distribution**, suggesting that market volatility arises from several distinct regimes rather than a single homogeneous process.

To capture this heterogeneous structure, we employ a **Gaussian Mixture Model (GMM)** estimated using the **Expectation–Maximization (EM)** algorithm. The GMM assumes that each observed VIX value x is generated from a mixture of K Gaussian components:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2),$$

where π_k denotes the weight of the k -th component ($\sum_k \pi_k = 1$), and each Gaussian distribution $\mathcal{N}(x \mid \mu_k, \sigma_k^2)$ represents a distinct volatility regime characterized by its own mean and variance. This probabilistic formulation enables us to uncover and interpret the **latent market states** underlying historical VIX data, offering a more flexible and realistic representation of volatility dynamics than a single Gaussian model.

The parameters of the Gaussian Mixture Model are estimated via the EM algorithm, which iteratively refines the estimates of the component weights and parameters. In the **E-step**, the posterior probability that each observation x_i belongs to component k is computed as

$$\gamma_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k^{(t)} \mathcal{N}(x_i \mid \mu_k^{(t)}, \sigma_k^{2(t)})}{\sum_{j=1}^K \pi_j^{(t)} \mathcal{N}(x_i \mid \mu_j^{(t)}, \sigma_j^{2(t)})}.$$

In the **M-step**, these posterior probabilities are used to update the parameters:

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \gamma_{ik}, \quad \mu_k^{(t+1)} = \frac{\sum_{i=1}^N \gamma_{ik} x_i}{\sum_{i=1}^N \gamma_{ik}}, \quad \sigma_k^{2(t+1)} = \frac{\sum_{i=1}^N \gamma_{ik} (x_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^N \gamma_{ik}}.$$

These two steps alternate until convergence, yielding maximum-likelihood estimates of the regime-specific parameters.

Limitations of GMM and Extensions

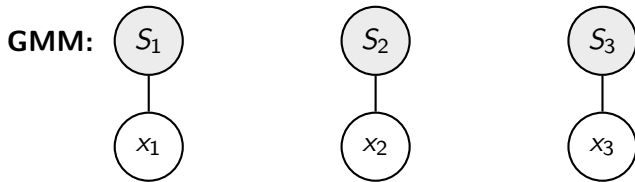
Limitation 1: No Temporal Dependence

While the Gaussian Mixture Model provides a probabilistic framework to identify multiple volatility regimes, it remains a static representation of market dynamics. The model assumes that the latent regime indicators z_t are independent across time, meaning each observation's state is determined solely by itself and not influenced by previous periods. This independence assumption prevents the GMM from capturing regime persistence or transition patterns that are fundamental to financial volatility dynamics. As a result, the Mixture Gaussian model is descriptive rather than predictive—it **characterizes the historical distribution of the VIX but cannot forecast or explain how market regimes evolve over time.**

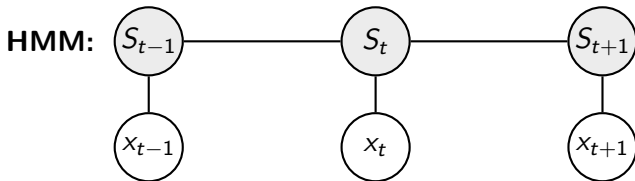
Solution: Hidden Markov Model (HMM)

To address these limitations, the Mixture Gaussian framework can be extended into dynamic regime-switching models that explicitly incorporate time dependence. A key example is the **Hidden Markov Model (HMM)**, where the latent regime follows a Markov process with defined transition probabilities between states. Unlike the static GMM, the HMM links each observation to both its current and previous states, allowing the model to capture regime persistence and transition dynamics. This time-dependent structure transforms the model from a purely descriptive tool into a **predictive framework, capable of estimating the likelihood of moving from a calm to a volatile market regime**, and thus offering deeper insight into the evolution of financial market risk.

GMM vs. HMM (Structure)



No temporal dependence among latent states



Add Markov property for the latent states

Limitation 2: Need to Predefine K

Another limitation is that we must predefine K , the number of mixture components. In the VIX example, we used AIC and BIC to select K , but this still requires trying multiple models and choosing the best one.

Solution: Dirichlet Process Gaussian Mixture (DP-GMM)

A more data-driven and flexible approach is to allow the model to infer the number of clusters automatically. Specifically, we can use a **Dirichlet Process Gaussian Mixture Model (DP-GMM)**, which is a nonparametric Bayesian method where the number of components can grow with the complexity of the data.